

Exact Hardy frequencies for quadratic parabolic Koenigs domains

FA/Banach run packet for arXiv:2507.08514

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Status. Candidate substantial partial result, likely valid. It completely resolves the explicit quadratic-boundary case singled out in arXiv:2507.08514v1, while the general parabolic Koenigs-domain problem remains open.

Source problem and exact scope

The source is C. Gómez-Cabello and F. J. González-Doña, *On frequencies of parabolic Koenigs domains*, arXiv:2507.08514v1 [1]. If $\Omega = h(\mathbb{D})$ is a Koenigs domain and $1 \leq p < \infty$, its H^p -frequencies are

$$\text{Freq}_p(\Omega) := \{\lambda \in \mathbb{C} : e^{\lambda z} \in H^p(\Omega)\}.$$

Equivalently, these are the eigenvalues of the infinitesimal generator of the associated composition semigroup on H^p .

Remark 5.14 on PDF page 38 of version 1 states that, for the convex domain with defining function $\psi_\Omega(t) = t^2$, the paper's estimates do not decide whether the two imaginary boundary rays contain frequencies:

$$\lambda = |\lambda|e^{i\pi/2}, \quad \mu = |\mu|e^{3i\pi/2}.$$

The current arXiv revision reorganizes this part and no longer prints this specific sentence. The packet therefore preserves version 1 as **source_paper.pdf**.

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ON FREQUENCIES OF PARABOLIC KOENIGS DOMAINS

Thus, for all $t \geq t_2$,

$$\omega(\zeta_t, F_t, S_t) \gtrsim \frac{1}{\psi_\Omega(-t)\pi^{\frac{\psi(-t)}{t}}} = \frac{1}{e^{\pi\frac{\psi(-t)}{t} \log \psi_\Omega(-t)}},$$

and the proof is done. $\square$

Remark 5.14. As Corollary 4.14 characterized $\sigma_p(\Delta|_{H^p})$ whenever $\psi(t) \geq C|t|^\gamma$ for $\gamma > 2$, the previous result also characterizes completely $\sigma_p(\Delta|_{H^p})$ under the convexity assumption on Ω and the growth condition $\psi(t) \leq C|t|^\gamma$ for $\gamma < 2$, $C > 0$ and $t \geq t_0$. Indeed, under such a hypothesis,

$$\sigma_p(\Delta|_{H^p}) = \left\{ re^{i\theta} : r \geq 0, \frac{\pi}{2} < \theta < \frac{3\pi}{2} \right\}.$$

It is worth remarking that, unlike the case in which Ω is convex and contains a sector, the point spectral picture of $\sigma_p(\Delta|_{H^p})$ is not completely characterized by Theorems 4.13 and 5.13. For instance, by considering $\psi_\Omega(t) = t^2$, it is straightforward to check that the aforementioned results do not provide any further information about the membership of $\lambda = |\lambda|e^{i\frac{\pi}{2}}$ and $\mu = |\mu|e^{i\frac{3\pi}{2}}$ to $\sigma_p(\Delta|_{H^p})$.

Proof intuition

The quadratic boundary is one of the rare non-sectorial Koenigs boundaries that can be uniformized exactly. The strip $\{|\operatorname{Im} \xi| < 1/2\}$, modulo the involution $\xi \mapsto -\xi$, is sent to the right of a parabola by $(\xi^2 + 1/4)/c$. The even map $\cosh(\pi\xi)$ descends through the same quotient and maps the parabolic domain conformally onto the right half-plane. Consequently the harmonic measure of the parabola is not merely estimated: it has the exact density $c \operatorname{sech}(\pi cy) dy$.

On the boundary, $|e^{\lambda z}|^p$ is a Gaussian-exponential factor $\exp(p(\operatorname{Re} \lambda)cy^2 - p(\operatorname{Im} \lambda)y)$. The Gaussian sign decides every non-imaginary frequency, while the exact exponential tail of harmonic measure gives the sharp imaginary threshold. An interior growth estimate after a Cayley transform proves the positive Hardy-membership direction without assuming an unproved boundary-value criterion.

Exact theorem

For $c > 0$, put

$$\Omega_c := \{x + iy \in \mathbb{C} : x > cy^2\}.$$

This is a convex simply connected domain satisfying $\Omega_c + t \subset \Omega_c$ for $t \geq 0$, hence a parabolic Koenigs domain.

Theorem 1. *For every $c > 0$ and $1 \leq p < \infty$,*

$$\operatorname{Freq}_p(\Omega_c) = \{\lambda \in \mathbb{C} : \operatorname{Re} \lambda < 0\} \cup \{it : t \in \mathbb{R}, |t| < \pi c/p\}.$$

In particular, the two critical frequencies $i\pi c/p$ and $-i\pi c/p$ do not belong to $\operatorname{Freq}_p(\Omega_c)$.

Proof. Step 1: exact uniformization. Let

$$S := \{\xi = u + iv : |v| < 1/2\}, \quad P_c(\xi) := \frac{\xi^2 + 1/4}{c}.$$

Writing $P_c(\xi) = x + iy$, a direct calculation gives

$$x - cy^2 = \frac{(1 - 4v^2)(u^2 + 1/4)}{c}.$$

Thus $P_c(S) = \Omega_c$. Every point of Ω_c has exactly the two preimages ξ and $-\xi$, except at the harmless branch point $\xi = 0$.

The map $\xi \mapsto \cosh(\pi\xi)$ sends S onto the right half-plane $\mathbb{H} := \{q : \operatorname{Re} q > 0\}$. Indeed,

$$\operatorname{Re} \cosh(\pi(u + iv)) = \cosh(\pi u) \cos(\pi v) > 0,$$

and the two preimages of a point are again ξ and $-\xi$. Therefore it descends to a conformal bijection

$$F_c : \Omega_c \longrightarrow \mathbb{H}, \quad F_c(z) = \cosh\left(\pi\sqrt{cz - \frac{1}{4}}\right).$$

The displayed expression is single-valued: its power series contains only integer powers of $cz - 1/4$. It also makes clear that the apparent branch at $z = 1/(4c)$ is removable.

Step 2: exact harmonic measure. The boundary is parameterized by

$$z(y) = cy^2 + iy, \quad y \in \mathbb{R}.$$

Since

$$cz(y) - \frac{1}{4} = (cy + i/2)^2,$$

we obtain

$$F_c(z(y)) = \cosh(\pi(cy + i/2)) = i \sinh(\pi cy).$$

Choose the base point $z_* = 1/(4c)$, for which $F_c(z_*) = 1$. Harmonic measure at 1 in the right half-plane has density

$$\frac{1}{\pi} \frac{dt}{1+t^2}$$

on the boundary it . Substituting $t = \sinh(\pi cy)$ gives the exact boundary density

$$d\omega_{\Omega_c, z_*}(z(y)) = c \operatorname{sech}(\pi cy) dy. \quad (1)$$

Step 3: the open left half-plane consists of frequencies. Write $\lambda = a + ib$. If $a < 0$, then for $z = x + iy \in \Omega_c$,

$$\operatorname{Re}(\lambda z) = ax - by \leq acy^2 - by.$$

The right-hand side is bounded above because its quadratic coefficient is negative. Hence $e^{\lambda z}$ is bounded on Ω_c , so it belongs to $H^\infty(\Omega_c) \subset H^p(\Omega_c)$.

Step 4: sufficiency on the imaginary axis. Let $\lambda = ib$, and let

$$Q(\zeta) = \frac{1+\zeta}{1-\zeta}, \quad \zeta \in \mathbb{D},$$

be the Cayley map to $\mathbb{H}$. Put $h_c = F_c^{-1} \circ Q$. For $q = Q(\zeta)$, choose $\xi = u + iv \in S$ with $h_c(\zeta) = (\xi^2 + 1/4)/c$. Then

$$|\operatorname{Im} h_c(\zeta)| = \frac{|2uv|}{c} \leq \frac{|u|}{c}.$$

Moreover,

$$|q|^2 = |\cosh(\pi(u + iv))|^2 = \sinh^2(\pi u) + \cos^2(\pi v),$$

so $\sinh(\pi|u|) \leq |q|$, and therefore

$$e^{\pi|u|} \leq 2|q| + 1 \leq \frac{C}{|1-\zeta|}$$

for an absolute constant C . It follows that

$$|e^{ibh_c(\zeta)}|^p \leq C_{b,c,p} |1-\zeta|^{-p|b|/(\pi c)}.$$

The standard disk estimate

$$\sup_{0 < r < 1} \int_{-\pi}^{\pi} |1 - re^{i\theta}|^{-\alpha} d\theta < \infty \iff \alpha < 1$$

now proves $e^{ibh_c} \in H^p(\mathbb{D})$ whenever $p|b| < \pi c$. By conformal invariance, $ib \in \operatorname{Freq}_p(\Omega_c)$.

Step 5: necessity and endpoint exclusion. If $e^{\lambda z} \in H^p(\Omega_c)$, its pullback by h_c is in $H^p(\mathbb{D})$, so its radial boundary values are p -integrable. Using conformal invariance of harmonic measure and (1), this forces

$$\int_{\mathbb{R}} \exp(p(acy^2 - by)) c \operatorname{sech}(\pi cy) dy < \infty. \quad (2)$$

If $a > 0$, the positive quadratic term makes (2) diverge. If $a = 0$, the asymptotic $\operatorname{sech}(\pi cy) \sim 2e^{-\pi c|y|}$ shows that the integral converges exactly when $p|b| < \pi c$. At $p|b| = \pi c$, the integrand tends to a positive constant along one tail, so both endpoints are excluded. Together with Steps 3 and 4, this proves the theorem. $\square$

Corollary 1 (all translated quadratic boundaries). *Let $c > 0$ and $x_0, y_0 \in \mathbb{R}$, and set*

$$\Omega_{c,x_0,y_0} := \{x + iy : x > x_0 + c(y - y_0)^2\}.$$

Then $\text{Freq}_p(\Omega_{c,x_0,y_0}) = \text{Freq}_p(\Omega_c)$, hence the formula in Theorem 1 remains valid.

Proof. The domain is the translate $x_0 + iy_0 + \Omega_c$. Translation multiplies $e^{\lambda z}$ by the nonzero constant $e^{\lambda(x_0 + iy_0)}$, which does not affect Hardy membership. $\square$

Connection to the source and limitations

For the source’s explicit example $\psi_\Omega(y) = y^2$, take $c = 1$. The theorem gives

$$\sigma_p(\Delta|_{H^p}) = \{\text{Re } \lambda < 0\} \cup \{it : |t| < \pi/p\}.$$

Thus each imaginary boundary ray contains precisely an open segment of frequencies, and the critical endpoint is absent. This completely answers the membership question posed by the example in arXiv:2507.08514v1.

The theorem does not characterize arbitrary convex parabolic Koenigs domains, nor defining functions that are merely asymptotic to a quadratic. Extending the result to that setting would require stable two-sided harmonic-measure asymptotics; the exact strip quotient used here is then unavailable.

Verification and upgrade attempts

1. The strip-to-parabola identity and boundary formula were checked symbolically and numerically by the included script `code/check_quadratic_map.py`. The script is only a regression check; the algebra in the proof is exact.
2. The initial $c = 1$ observation was upgraded to every $c > 0$, and then to all translated quadratic boundaries. This changes the critical value from π/p to $\pi c/p$.
3. The positive endpoint direction was checked independently using an interior disk-growth estimate, rather than relying only on boundary integrability.
4. An upgrade to all asymptotically quadratic defining functions was considered. The proof does not supply the required harmonic-measure stability, so no such claim is made.

Novelty check

Before promotion, the run indexes were searched for arXiv:2507.08514, parabolic Koenigs frequencies, quadratic defining functions, imaginary boundary frequencies, and related frequency-completeness work. Bounded web/arXiv searches through 17 August 2026 used the exact source title and close variants involving $\psi(t) = t^2$, $x > y^2$, boundary frequencies, and Hardy exponentials. They found the source, its conference abstract, and adjacent frequency-completeness literature, but no prior exact uniformization or the frequency formula proved here. Novelty confidence is moderate: this is an elementary explicit-domain computation that may exist outside the searched terminology.

Human-review recommendation

Review as a likely valid substantial partial result. The key checks are: (i) that the even strip maps descend to a conformal bijection; (ii) the harmonic-measure change of variables; and (iii) the uniform H^p estimate at the disk point corresponding to infinity. The full general parabolic-domain problem is not claimed solved.

References

- [1] C. Gómez-Cabello and F. J. González-Doña, *On frequencies of parabolic Koenigs domains*, arXiv:2507.08514v1, 2025.
- [2] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970.